🤖 IntellMeet – AI-Powered Enterprise Meeting & Collaboration Platform

Real-Time Video Meetings • AI Summaries • Smart Action Items • Team Collaboration

Production-Grade MERN Full-Stack System with AI Intelligence Zidio Development / Portfolio / Interview Reference April 2026

Title: IntellMeet – AI-Powered Enterprise Meeting & Collaboration Platform
subtitle: Production-Grade Full-Stack MERN Application with Real-Time Video, AI Meeting Intelligence & Team Collaboration
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"Meetings are the biggest time killer in enterprises. IntellMeet transforms meetings into productive experiences with real-time video, AI-powered summaries, smart action item extraction, and seamless collaboration — reducing meeting follow-up time by 40–60%."

1. Business Case & Production Objectives (Heavy Bit-by-Bit Explanation)

Mission
Build a production-grade enterprise meeting and collaboration platform using the MERN stack that combines real-time video meetings, AI-powered meeting intelligence (transcription, summary, action items), team chat, task management, and analytics — designed for modern remote/hybrid teams at Zidio Development's enterprise clients.

Why This Project Matters for Zidio Development
Enterprises waste thousands of hours every year in unproductive meetings. IntellMeet solves this by turning every meeting into an actionable, trackable event. It is perfect for Zidio's enterprise clients who need reliable, scalable, and intelligent collaboration tools.

Quantified Business Impact Targets (Bit-by-Bit)

- Reduce meeting follow-up time by **40–60%** through AI summaries and automatic action items — employees spend less time writing follow-up emails and chasing tasks.
- Improve team productivity by **25–40%** with real-time collaboration and task tracking — meetings become actionable instead of forgotten.

- Support **500–5,000 concurrent meeting participants** — the platform must handle large company town halls or cross-team meetings without lag.
- Achieve **99.95%** uptime SLA — critical for business-critical meetings.
- Zero-downtime deployments — new AI features can be released without interrupting ongoing meetings.

Non-Functional Requirements (Heavy Bit-by-Bit Explanation)

- Latency: < 200 ms for real-time features (video, chat, AI updates) — users expect instant response during meetings.
- Throughput: Handle 10k+ concurrent meetings — the system must scale during peak hours (Monday mornings, quarterly reviews).
- Availability: 99.95% — meetings cannot fail due to downtime.
- Security: End-to-end encryption for meetings, JWT authentication, role-based access — sensitive business discussions must remain private.
- Scalability: Horizontal scaling with Socket.io clustering and Kubernetes — add servers automatically when user load increases.
- Observability: Full logging, error tracking, performance monitoring — know exactly what is happening at any time for quick debugging.

2. Core Functional Requirements (with Heavy Bit-by-Bit Explanation)

ID	Capability	Detailed Description & Business Value	Key Acceptance Criteria & Production Metrics
F-01	User Authentication & Profiles	Secure signup/login with JWT, profile creation, team invitation	OAuth2 support, password hashing, role-based access (Admin, Member)
F-02	Real-Time Video Meetings	Video conferencing with screen sharing, recording, and live transcription	Support 50+ participants per meeting, low latency
F-03	AI Meeting Intelligence	Automatic transcription, summary generation, action item extraction using AI	Accurate summaries with >85% accuracy, action items with assignees
F-04	Real-Time Chat & Collaboration	In-meeting chat, shared notes, task creation during meeting	Real-time sync across all participants
F-05	Post-Meeting Dashboard	Meeting history, recordings, summaries, action items tracking	Searchable history, export options

ID	Capability	Detailed Description & Business Value	Key Acceptance Criteria & Production Metrics
F-06	Team & Project Management	Team workspaces, project boards, task assignment	Kanban-style boards with real-time updates
F-07	Analytics & Insights	Meeting frequency, productivity metrics, engagement reports	Dashboard with charts and exportable reports

Heavy Bit-by-Bit Explanation

The AI Meeting Intelligence feature uses OpenAI or Hugging Face models to transcribe the meeting in real time, generate a concise summary, and extract action items with assigned owners. This saves hours of manual note-taking and follow-up.

3. Production Technology Stack – 2026 (with Heavy Bit-by-Bit Explanation)

Layer	Primary Technology	Rationale / Alternatives
Frontend	React 19 + TypeScript + Vite	Fast HMR, code-splitting
UI Components	shadcn/ui + Tailwind CSS v4	Modern, accessible, customizable
State Management	TanStack Query + Zustand	Server-state + lightweight client-state
Backend	Node.js + Express	Lightweight, fast, scalable
Database	MongoDB (with Mongoose)	Flexible schema for meetings and tasks
Real-Time	Socket.io + WebRTC	Bidirectional real-time communication + video
AI Integration	OpenAI / Hugging Face (for summarization)	AI-powered transcription and action item extraction
Cache	Redis	Session management and feed caching
Authentication	JWT + bcrypt	Secure, stateless authentication
File Storage	Cloudinary / AWS S3	Scalable media storage for recordings
Containerization	Docker multi-stage	Consistent, lightweight deployments
Orchestration	Kubernetes + Helm	Auto-scaling and high availability
CI/CD	GitHub Actions	Automated testing and deployment
Monitoring	Prometheus + Grafana + Sentry	Full observability

Heavy Bit-by-Bit Explanation
Socket.io + WebRTC is used for real-time video and chat because it provides low-latency bidirectional communication. OpenAI is integrated for AI features because it delivers high-quality transcription and summarization out of the box.

4. 28-Day Day-by-Day Execution Plan (Pin-to-Pin Ultra Heavy Detailing with Bit-by-Bit Explanation)

Week 1 – Core Backend & Authentication Foundation

Day 1
• Project setup: MERN boilerplate with Vite for frontend and Express for backend
• MongoDB connection and basic server configuration
• Install all core dependencies (express, mongoose, dotenv, cors, helmet, socket.io, web-rtc)
• Create folder structure and initial git commit
• **Bit-by-Bit Explanation**: This day establishes the foundation so all future features can be built on a stable, secure base. Proper folder structure prevents technical debt later.

Day 2
• User model and authentication routes (signup, login)
• JWT implementation with refresh tokens
• Implement password hashing with bcrypt
• **Bit-by-Bit Explanation**: JWT is used because it is stateless and secure for API authentication. Refresh tokens improve user experience by reducing frequent logins.

Day 3
• Profile creation and avatar upload using Cloudinary
• Protected routes with middleware
• Add rate limiting on auth routes using express-rate-limit
• **Bit-by-Bit Explanation**: Cloudinary is chosen for media upload because it handles resizing and CDN delivery automatically. Rate limiting prevents brute-force attacks.

Day 4
• Meeting model and basic CRUD for meetings
• WebRTC setup for video calls (peer connection logic)
• **Bit-by-Bit Explanation**: WebRTC is the industry standard for peer-to-peer video conferencing. Basic CRUD ensures meetings can be created and managed from day one.

Day 5
• Redis setup for session and meeting caching
• Socket.io server configuration for real-time features

• **Bit-by-Bit Explanation**: Redis ensures fast access to session data and cached meetings. Socket.io is configured early for real-time capabilities.

Day 6

• Basic chat functionality in meetings

• Real-time notification setup using Socket.io events

• **Bit-by-Bit Explanation**: Real-time notifications keep users engaged by showing activity instantly. Chat is implemented as a core feature for collaboration.

Day 7

• Week 1 checkpoint: Backend API running locally, authentication working, basic meeting creation and real-time connection tested with Postman

• Write initial README and commit all changes

• **Bit-by-Bit Explanation**: This checkpoint ensures the core foundation is solid before moving to frontend work. Early documentation helps maintain clarity.

Week 2 – Frontend & Real-Time Meeting Core

Day 8

• React 19 app setup with TypeScript, shadcn/ui, Tailwind, TanStack Query, Zustand

• Install necessary packages for video and real-time features

Day 9

• Authentication pages and protected routes

Day 10

• Meeting lobby and video room UI with WebRTC integration

Day 11

• Real-time chat inside meeting room with typing indicators

Day 12

• Screen sharing and recording controls

Day 13

• Live participant list with presence indicators and mute controls

Day 14

• Week 2 checkpoint: Real-time video meeting and chat fully functional

• Test end-to-end meeting creation and joining

Week 3 – AI Intelligence & Collaboration Features

Day 15

• AI transcription integration (OpenAI Whisper or similar)

Day 16

• AI meeting summary and action item extraction

Day 17

• Post-meeting dashboard with summaries and action items

Day 18

• Team workspace and project board (Kanban style)

Day 19

• Task creation from meeting action items with assignee selection

Day 20

• Notification system for mentions and action items

Day 21

• Week 3 checkpoint: AI features and collaboration tools operational

• Test AI summary accuracy on sample meetings

Week 4 – Deployment, Monitoring & Production Polish

Day 22

• Docker multi-stage builds for frontend and backend

Day 23

• Kubernetes manifests and Helm chart

Day 24

• GitHub Actions CI/CD pipeline with testing and deployment stages

Day 25

• Cloud deployment (AWS or Vercel + Render) with environment variables setup

Day 26

• Prometheus + Grafana monitoring and Sentry error tracking

Day 27

• Load testing with JMeter and final security review with OWASP ZAP

Day 28

• Final QA, edge case testing, README polishing, demo video recording, PDF export

5. Challenges, Learnings & Industry Best Practices

- Real-time video scaling handled with WebRTC + Socket.io
- AI integration for meeting intelligence
- Best practices: Event-driven architecture, CI/CD, observability

6. Security & Privacy Highlights

- JWT with refresh tokens
- OWASP Top 10 mitigation
- End-to-end encryption for meetings (optional)
- Rate limiting
- Secrets managed securely

Zidio Development – Web Development Domain

Project Submission Guidelines
March 2026 Edition

Prepared for participants
Focus: Industry-grade full-stack / advanced web projects
Submission period: You need to submit the project on or before the due date if you submit after the due date means you will be automatically disqualified for stipend by the system.
Evaluation emphasis: Code quality · Documentation · Live demo · Scalability & security awareness

1. General Rules & Eligibility

- All code must be **original work** created mainly during the LogicVeda preparation/submission window.
- Participants must submit **exactly 1 project** (as per the planned structure: 1 month each, weekly breakdown).
- Projects should demonstrate **progressive complexity**:
    1. Strong modern frontend / real-time foundations
    2. Distributed systems / microservices / resilience

3. AI/ML-integrated or data-heavy web application

- Plagiarism / Ai generated content will result in disqualification for stipend.

2. Mandatory Submission Deliverables

Submit **one consolidated package** containing all three projects:

#	Deliverable	Format / Location	Required?	Evaluation Weight
1	Project Documentation / Report (1 PDF)	1 PDF file (A4, 8–15 pages)	Yes	25%
2	Live Public Demo URL	1 HTTPS link (no mandatory login)	Yes	30%
3	GitHub Repository	1 repo with clear folders	Yes	20%
4	README.md per project	Detailed, professional README in repo	Yes	15%
5	Demo Video(s)	3–7 min per project (LinkedIn Post/ YouTube unlisted / Loom / Drive)	Yes	10%

Naming convention recommendation
`AI-Powered Enterprise Meeting & Collaboration_Platform_Zidio_March2026.pdf/zip`

3. Documentation Guidelines – What every PDF should contain

Use consistent structure across all three documents for a polished portfolio feel.

1. **Hero / Cover Section**
    - Large project title
    - Catchy tagline
    - Your name + @harsadash + date
    - Gradient background (optional but recommended)
2. **Project Overview**
    - Vision & objectives
    - Target users / use cases
    - Business value delivered
    - Non-functional goals (latency, concurrency, availability targets)

3. **Key Features** (table format)
   | ID | Feature | Description | Acceptance Criteria |

4. **Technology Stack** (table)
   | Category | Technology | Rationale / Alternatives |

5. **Architecture Diagram**

   - Include screenshot (Excalidraw, Draw.io, Lucidchart, etc.)
   - Or clean ASCII art if no image tool used

6. **Detailed Execution Timeline**

   - Day-by-day or week-by-week breakdown
   - Major deliverables per phase
   - Milestones & checkpoints

7. **Technical Highlights**

   - Security measures (OWASP mitigation, input sanitization, rate limiting, etc.)
   - Performance & scalability notes (caching, lazy loading, load test results)
   - Challenges faced & how solved

8. **Deployment & Operations**

   - Platform used (Vercel, Render, Railway, AWS, Fly.io, etc.)
   - CI/CD pipeline summary
   - Monitoring / health checks (if implemented)

9. **Visuals**

   - 5–10 high-quality screenshots / GIFs
   - Architecture diagram
   - Live demo captures

10. **Personal Reflection** (optional but strongly recommended)

    - Key learnings
    - Industry best practices applied
    - Future roadmap ideas

4. Code & Repository Expectations

- Clean, modular code structure
- Consistent naming & formatting (use ESLint / Prettier)
- Semantic commit messages (`feat: add real-time cursor presence` , `fix: JWT refresh token bug` , etc.)
- Feature branches & pull requests (even if solo)
- `.gitignore` properly configured
- No committed secrets / API keys
- Basic tests appreciated (even 30–50% coverage shows intent)
- Responsive design + accessibility basics (ARIA labels, keyboard navigation)

5. Live Demo Requirements

- Publicly accessible (no VPN / geo-restriction)
- HTTPS enforced
- Core functionality available without sign-up (demo account OK if needed)
- Fast initial load (< 5 seconds preferred)
- Mobile responsive (test on phone / tablet)
- Include brief instructions on demo page if non-obvious

6. Evaluation Criteria (Suggested 100-point scale)

Category	Points	Focus Areas
Innovation & Problem Solving	15	Original features, thoughtful design choices
Technical Depth & Best Practices	25	Clean code, modern stack, security & performance awareness
Functionality & User Experience	20	Works as promised, intuitive UX, responsive
Documentation Quality	20	Clear, professional, well-structured
Deployment & Reliability	10	Stable live demo, easy local setup
Presentation & Polish	10	Demo video quality, visual appeal of docs

7. Important Rules & Tips

- **Strict deadline** – no late submissions accepted

- **File size limits**: ZIP ≤ 500 MB, individual PDFs ≤ 10 MB
- **No external dependencies that require payment / private keys** for judges to run demo
- **Backup plan**: If live demo is down, high-quality video + screenshots must still convey full functionality
- **Stand-out practices**:
  - Include Lighthouse scores / performance metrics
  - Show real multi-user session in video (especially Project 1)
  - Mention OWASP mitigations, load test results, CI/CD pipeline
  - End docs with personal growth reflection

Submission email / form subject example

`Thers no email submission you need to submit the project in dashboard itself`

Good luck!
Submit with confidence — your detailed planning and advanced topics already make these projects stand out.

Team LogicVeda
March 2026

Crafted with precision and modern engineering principles • LogicVeda Technologies • March 2026